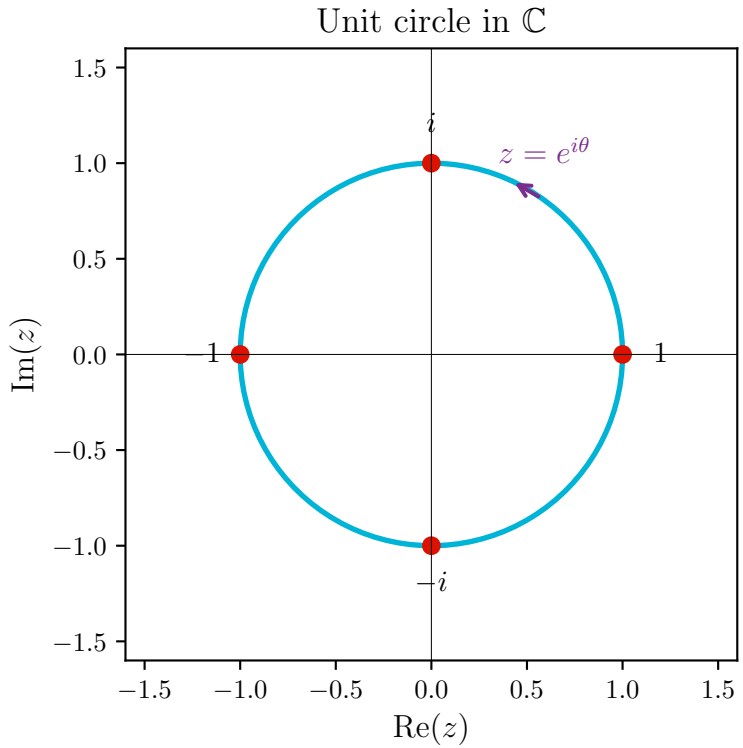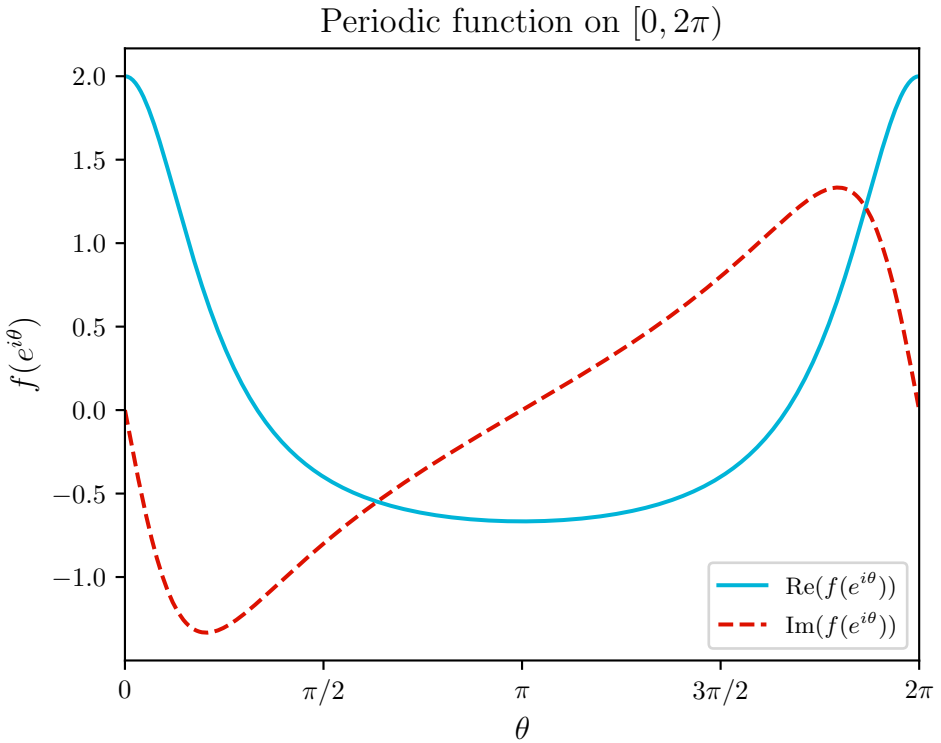$z = e^{i\theta}$: Laurent series $\longrightarrow$ Fourier series